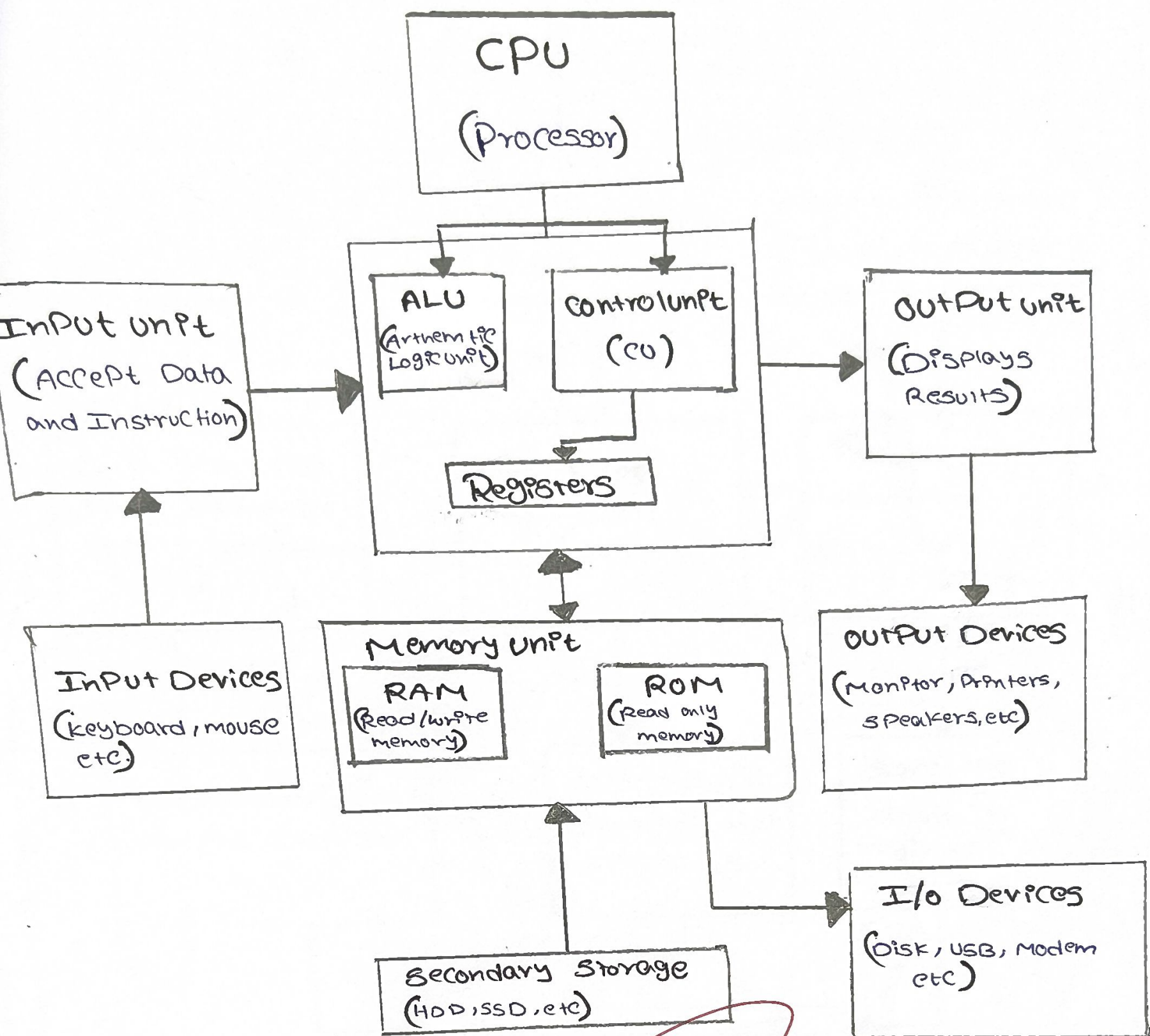
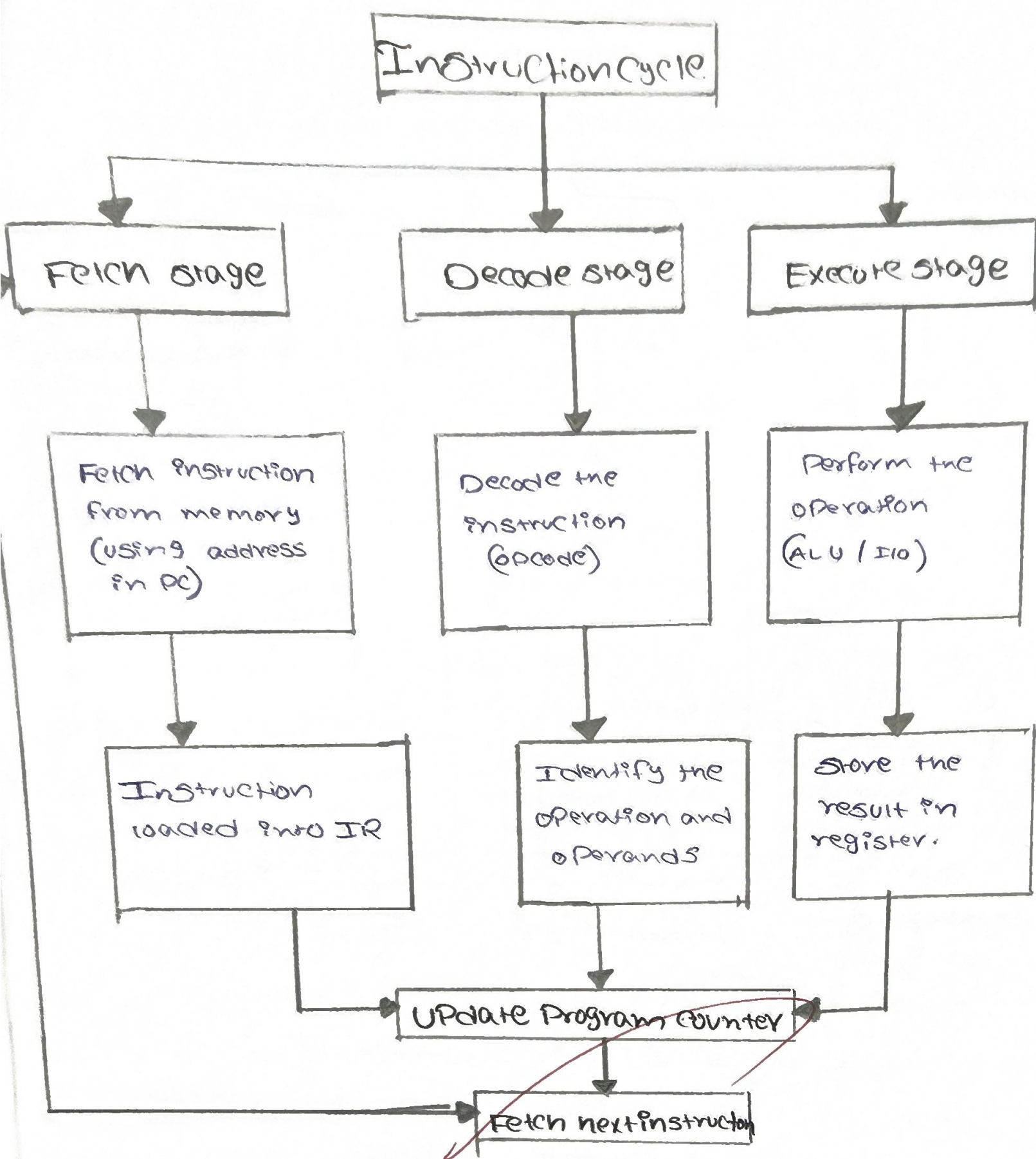
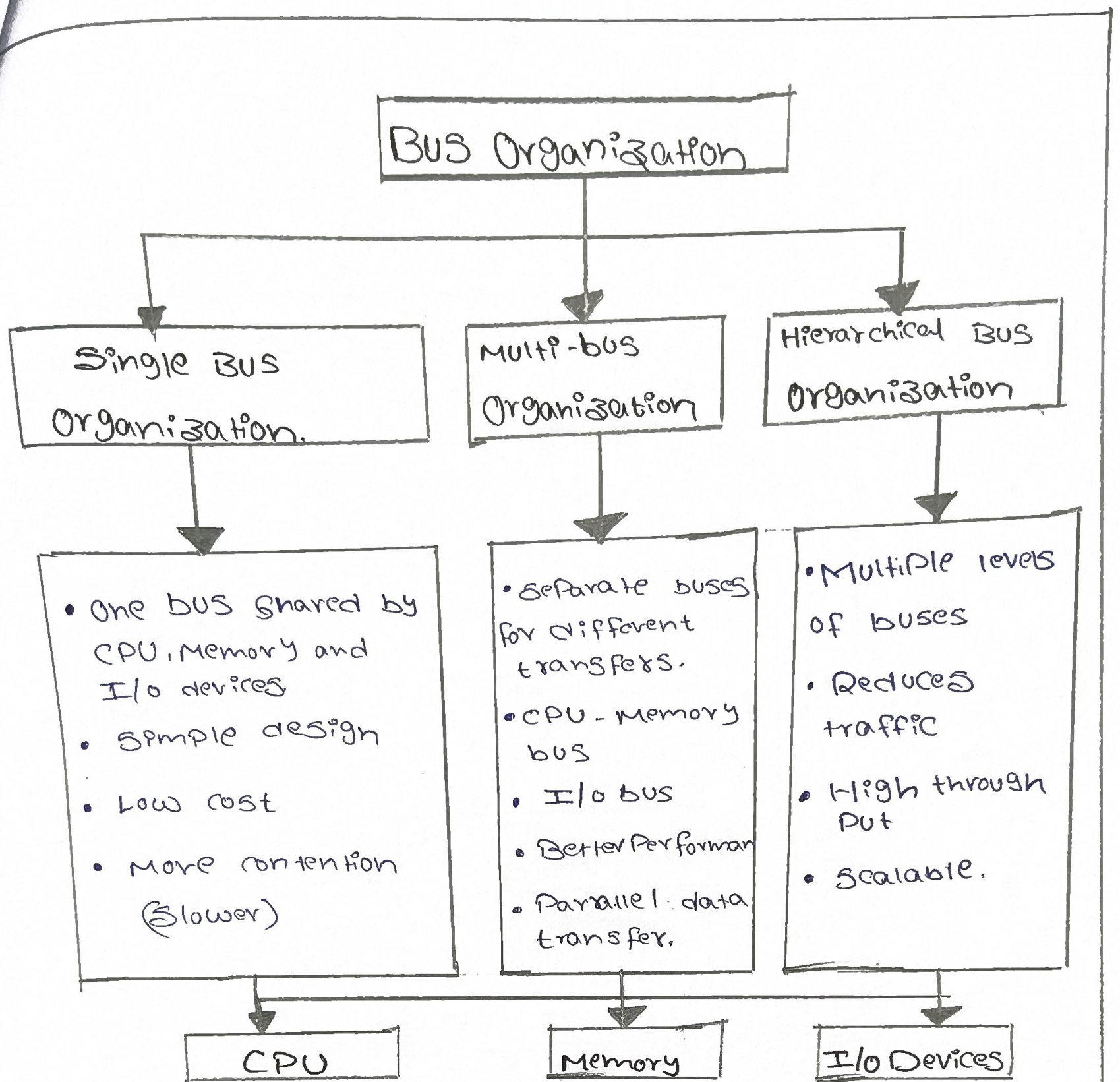




The Input unit accepts data, the Memory unit stores instructions and data, the CPU fetches and executes instructions using the ALU and control unit, and the output displays the results. All functional units work together to execute programs efficiently.

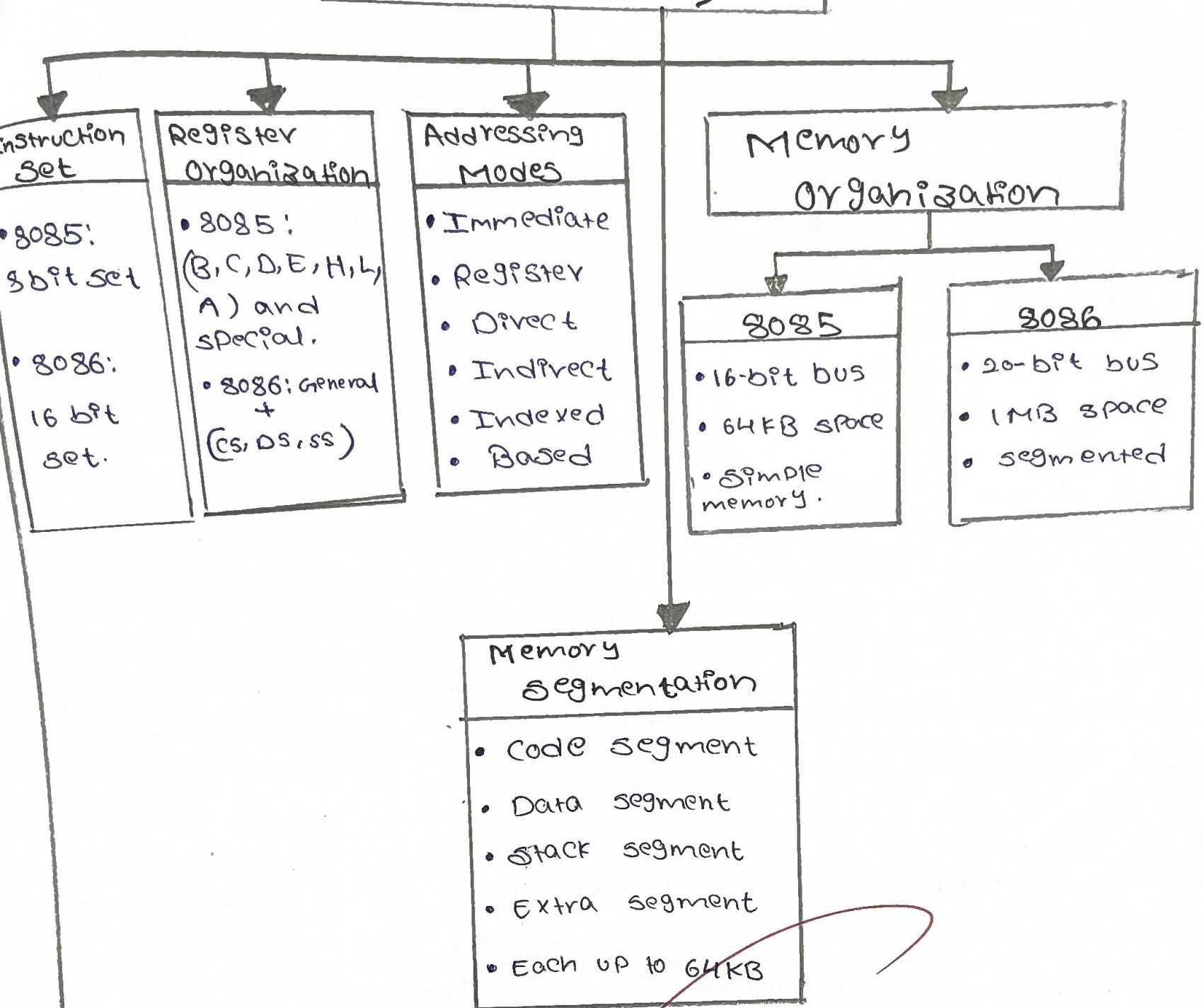


The fetch stage reads the instruction from memory the Decode stage identifies the operation and operands. and the execute stage performs the required operation. These stages ensure correct and efficient program execution.



Single BUS uses one shared bus for all devices, Multi-bus uses separate buses to increase data transfer speed, and Hierarchical BUS uses multiple bus levels to reduce traffic and improve communication efficiency.

Microprocessor Architectures (8085 and 8086)



The 8085 uses a simple 64KB memory organization. While the 8086 uses segmented memory, allowing access to 1MB of memory. The 8086 has more registers and addressing modes, resulting in faster instruction execution and better memory management.

Number System

Binary

- Base-2 (0 & 1)
- Used in digital computers
- Machine language

Hexadecimal

- Base-16 (0-9, A-F)
- Compact form of binary
- Easy to read and write

Conversions

- Binary to Hexadecimal
- Used in assemblers and debuggers

Ins. Representation

- Instructions in machine code represented in Hexadecimal

Memory Storage

- Instructions and data stored in binary form.
- Memory addresses in hexadecimal

CPU Interpretation

- Reads binary from memory
- Executes instructions.

Program execution & Debugging

- Efficient execution
- Easy debugging
- Better memory organization

Binary is the language understood by the CPU, while hexadecimal provides a shorter and easier representation of binary values. Together, they simplify instruction and debugging.